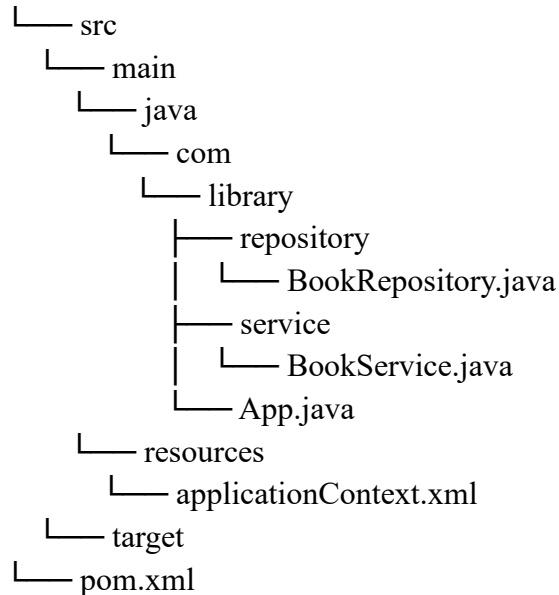


Module 5: Spring Core and Maven

1. Configuring a basic Spring Application

The Project directory:

LIBRARYMANAGEMENT



Codes:

Book Repository.java

```
package com.library.repository;
public class BookRepository {
    public BookRepository() {
        System.out.println("BookRepository bean created.");
    }
    public void getBookDetails() {
        System.out.println("Fetching book details from the repository...");
    }
}
```

BookService.java

```
package com.library.service;
import com.library.repository.BookRepository;
public class BookService {
    private BookRepository bookRepository;
    public BookService() {
        System.out.println("BookService bean created.");
    }
    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }
    public void displayBook() {
```

```

        System.out.println("BookService: Processing request...");
        bookRepository.getBookDetails();
    }
}

```

App.java

```

package com.library;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;
public class App {
    public static void main(String[] args) {
        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");
        BookService service =
            context.getBean("bookService", BookService.class);
        service.displayBook();
    }
}

```

applicationContext.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
        https://www.springframework.org/schema/beans/spring-beans.xsd">
    <bean id="bookRepository"
        class="com.library.repository.BookRepository"/>
    <bean id="bookService"
        class="com.library.service.BookService">
        <property name="bookRepository"
            ref="bookRepository"/>
    </bean>
</beans>

```

pom.xml

```

<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
        https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.library</groupId>
    <artifactId>LibraryManagement</artifactId>
    <version>1.0</version>
    <packaging>jar</packaging>
    <properties>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
        <maven.compiler.source>21</maven.compiler.source>
        <maven.compiler.target>21</maven.compiler.target>
    </properties>

```

```

</properties>
<dependencies>
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-context</artifactId>
    <version>5.3.39</version>
  </dependency>
</dependencies>
<build>
  <plugins>
    <plugin>
      <groupId>org.codehaus.mojo</groupId>
      <artifactId>exec-maven-plugin</artifactId>
      <version>3.6.0</version>
      <configuration>
        <mainClass>com.library.App</mainClass>
      </configuration>
    </plugin>
  </plugins>
</build>
</project>

```

Output:

```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- exec:3.6.0:java (default-cli) @ LibraryManagement ---
BookRepository bean created.
BookService bean created.
BookService: Processing request...
Fetching book details from the repository...
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.417 s
[INFO] Finished at: 2026-07-25T20:23:21+05:30
[INFO] -----

```

2. Implementing Dependency Injection

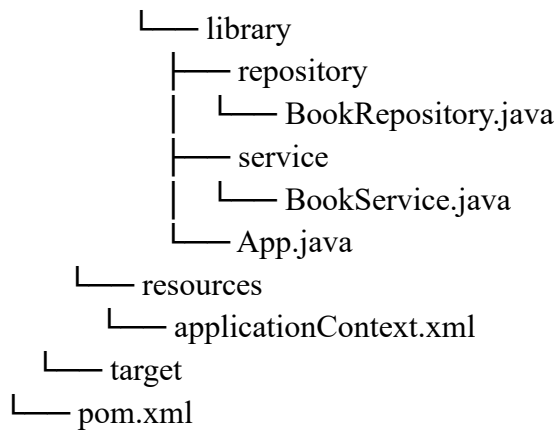
The Project directory:

LIBRARYMANAGEMENT

```

└─ src
    └─ main
        └─ java
            └─ com

```



Codes:

BookRepository.java

```
package com.library.repository;
public class BookRepository {
    public BookRepository() {
        System.out.println("BookRepository bean created.");
    }
    public void getBookDetails() {
        System.out.println("Book Name : Spring Framework");
        System.out.println("Author      : Rod Johnson");
        System.out.println("Status       : Available");
    }
}
```

BookService.java

```
package com.library.service;
import com.library.repository.BookRepository;
public class BookService {
    private BookRepository bookRepository;
    public BookService() {
        System.out.println("BookService bean created.");
    }
    // Setter Injection
    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }
    public void displayBookDetails() {
        System.out.println("Library Management System");
        System.out.println("-----");
        bookRepository.getBookDetails();
    }
}
```

App.java

```
package com.library;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
```

```
import com.library.service.BookService;
public class App {
    public static void main(String[] args) {
        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");
        BookService service =
            context.getBean("bookService", BookService.class);
        service.displayBookDetails();
    }
}
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
        https://www.springframework.org/schema/beans/spring-beans.xsd">
    <!-- Repository Bean -->
    <bean id="bookRepository"
        class="com.library.repository.BookRepository"/>
    <!-- Service Bean -->
    <bean id="bookService"
        class="com.library.service.BookService">
        <property
            name="bookRepository"
            ref="bookRepository"/>
    </bean>
</beans>
```

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
        https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.library</groupId>
    <artifactId>LibraryManagement</artifactId>
    <version>1.0</version>
    <packaging>jar</packaging>
    <properties>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
        <maven.compiler.source>21</maven.compiler.source>
        <maven.compiler.target>21</maven.compiler.target>
    </properties>
    <dependencies>
        <dependency>
            <groupId>org.springframework</groupId>
            <artifactId>spring-context</artifactId>
```

```

        <version>5.3.39</version>
    </dependency>
</dependencies>
<build>
<plugins>
    <plugin>
        <groupId>org.codehaus.mojo</groupId>
        <artifactId>exec-maven-plugin</artifactId>
        <version>3.6.0</version>
        <configuration>
            <mainClass>com.library.App</mainClass>
        </configuration>
    </plugin>
</plugins>
</build>
</project>

```

Output:

```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

[INFO]
[INFO] --- exec:3.6.0:java (default-cli) @ LibraryManagement ---
BookRepository bean created.
BookService bean created.
Library Management System
-----
Book Name : Spring Framework
Author    : Rod Johnson
Status    : Available
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time:  1.171 s
[INFO] Finished at: 2026-07-25T21:29:05+05:30
[INFO] -----

```

4. Creating and Configuring a Maven Project

Codes:

pom.xml

```

<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
        http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.library</groupId>
    <artifactId>LibraryManagement</artifactId>
    <version>1.0</version>

```

```
<packaging>jar</packaging>
<properties>
  <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  <maven.compiler.source>21</maven.compiler.source>
  <maven.compiler.target>21</maven.compiler.target>
</properties>
<dependencies>
<!-- Spring Context -->
<dependency>
  <groupId>org.springframework</groupId>
  <artifactId>spring-context</artifactId>
  <version>5.3.39</version>
</dependency>
<!-- Spring AOP -->
<dependency>
  <groupId>org.springframework</groupId>
  <artifactId>spring-aop</artifactId>
  <version>5.3.39</version>
</dependency>
<!-- Spring Web MVC -->
<dependency>
  <groupId>org.springframework</groupId>
  <artifactId>spring-webmvc</artifactId>
  <version>5.3.39</version>
</dependency>
</dependencies>
<build>
<plugins>
  <plugin>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-compiler-plugin</artifactId>
    <version>3.11.0</version>
    <configuration>
      <source>1.8</source>
      <target>1.8</target>
    </configuration>
  </plugin>
</plugins>
</build>
</project>
```

Output:

```
PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ LibraryManagement ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 3 source files with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path not set in conjunction with -source 8
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.105 s
[INFO] Finished at: 2026-07-25T21:52:44+05:30
[INFO] -----
```